LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 7: Strings: Working with Text

Module 3 | 45 minutes | String processing | Grades 6–8 · ages 12–14

Objective: Treat text as a sequence of characters: loop through letters, count matches, reverse a word and test for palindromes.

Success criteria

- I can loop through the letters of a string.
- I can count and build strings.
- I can test for palindromes.

Key vocabulary

- **string**: A sequence of characters (text).
- **character**: One letter, digit, space or symbol.
- **concatenate**: Join strings together.
- **palindrome**: Reads the same forwards and backwards.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Letter tiles (5 min): learners reverse a word by moving tiles one at a time to the front of a new row.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Vowel counter and reverser: learners follow the steps.
14 min	**Perform**	Code Lab: count the vowels, Code Lab: palindrome checker, Trace the reverse
9 min	**Probe**	Strings in Python, activity
4 min	**Wrap-up**	Exit ticket: Which block gives the third letter of a word?

Unplugged warm-up

Letter tiles (5 min): learners reverse a word by moving tiles one at a time to the front of a new row.

Teaching the idea

Strings are indexable sequences (1-based in Scratch, 0-based in Python). Traversal with letter (i) of, accumulation with join (prepending reverses), membership tests via a list, and comparison for palindromes.

Analogy: Beads on a string: each bead (character) has a position.

Live demo: Trace reversing “code” on the board: c, oc, doc, edoc.

Questions to ask

Ask	Listen for
What does letter 3 of [apple] give?	p.
Why does joining to the front reverse?	Each new letter becomes first.
How do you test a palindrome?	Compare with its reverse.

Differentiation

Support
Provide the loop; learners write the body.

Stretch
Ignore spaces so “never odd or even” counts as a palindrome.

Common misconceptions

Misconception	How to fix it
join (rev) (ch) reverses	That rebuilds the word in the same order.
Strings start at 0	In Scratch they start at 1; Python at 0.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which block gives the third letter of a word?

Homework
Find five palindromes (words or numbers) and test them with your program.

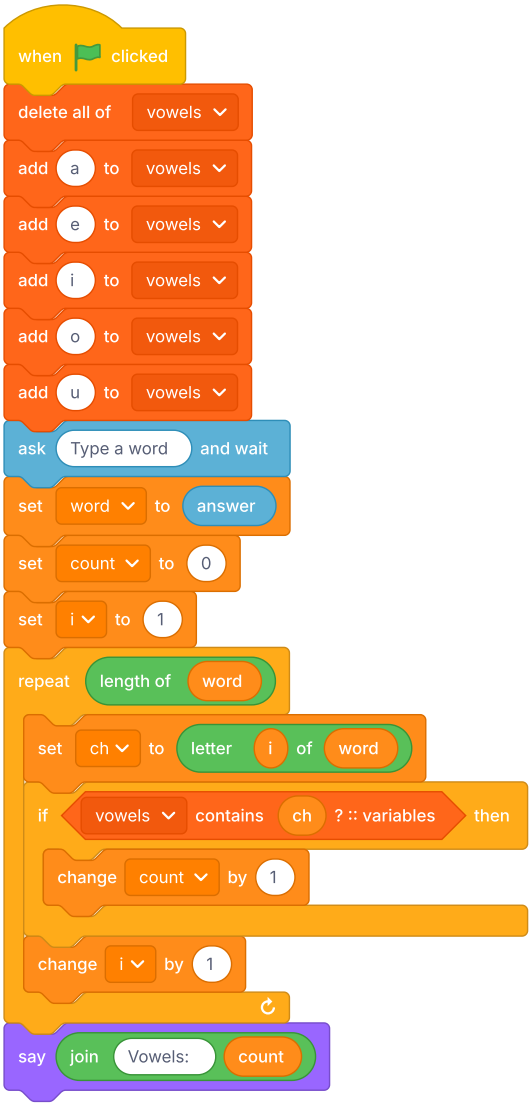
Answer key — Kitabash activities

1. Code Lab: count the vowels (perform · code lab)

Goals checked: banana → 3 · rhythm → 0 · Uses letter of

One solution:

Agent

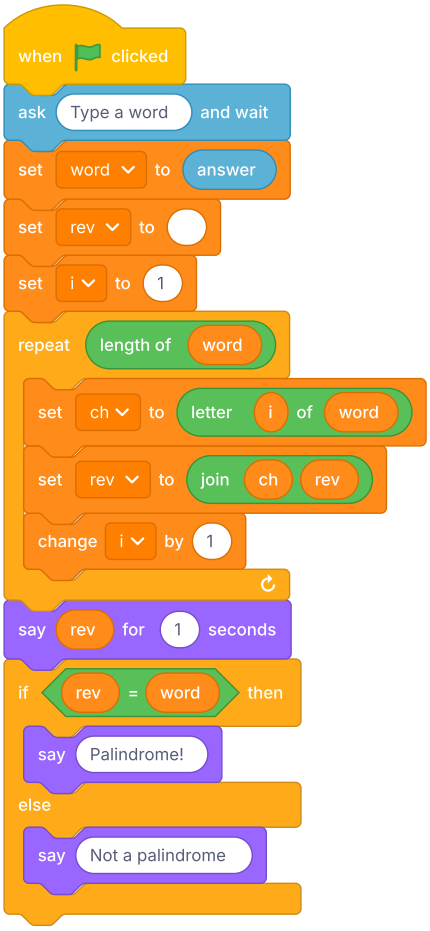


2. Code Lab: palindrome checker (perform · code lab)

Goals checked: racecar is a palindrome · apple is not · Builds the reverse

One solution:

Agent



3. Trace the reverse (perform · multiple choice)

Answer: edoc

4. Strings in Python (probe · code output)

5. Explain your thinking (probe · free response)

Look for: letter, text, check. **Example:** A password checker goes through the password letter by letter. It checks whether there is at least one number and one capital letter, and counts the length, then tells you if the password is strong enough.

Answer key — worksheet

1. **Key words:** string = A sequence of characters (text). · character = One letter, digit, space or symbol. · concatenate = Join strings together. · palindrome = Reads the same forwards and

backwards.

2. **Plan your code:** Goals: banana → 3; rhythm → 0; Uses letter of.
3. **Plan your code:** Goals: racecar is a palindrome; apple is not; Builds the reverse.
4. **Trace the reverse:** A. edoc — Each new letter goes in front: c → oc → doc → edoc.
5. **Strings in Python:** A. True then 5

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.